

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

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Abstract

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1 Introduction

In studying the evolving nature of our universe on the largest scales, galaxy clusters represent the most expansive clearly defined structures in which matter approaches virialization. Due to this fact, we can use them as laboratories to study the cosmic growth rate and characterize density fluctuations, as well as reconstruct the dynamical history of merging systems and the effects of feedback from active members. A tenuous plasma called the Intracluster Medium (ICM) permeates these massive objects, with temperatures in the 10^7 K range and densities from 10^{-5} to 10^{-1} cm^{-3} between their outer regions and inner cores. In this hot state, the ICM radiates at several keV photon energies, making spectroscopy in the soft X-ray band a primary method to resolve the distribution of baryonic matter inside clusters (Böhringer & Werner 2010).

1.1 Models

At the broadest level, galaxy clusters are modeled as dark matter halos fully governed by gravitational collapse. They are thought to originate from primordial density fluctuations and grow via cluster mergers as well as by accreting matter from cosmic filaments. Their total mass is dominated by the dark component defined through its purely gravitational interactions, the density of which is well described using a Navarro-Frenk-White (NFW) profile. Most baryonic matter is not contained in the stellar mass of the member galaxies themselves but in the ICM gas, which follows a similar shape to the overall matter distribution in the outer parts where it usually makes up a fraction of roughly $f_g \sim 15\%$ depending on redshift, but flattens towards the central region due to counteracting pressures. Containing this material requires a sufficiently deep gravitational potential well that is provided by the high dark matter concentration, which strongly peaks in the center as collisional effects are negligible. Shown in **Figure 1** are exemplary radial density profiles with realistic parameter choices for the observed galaxy cluster categories as explained in the following subsections (Pratt et al. 2019).

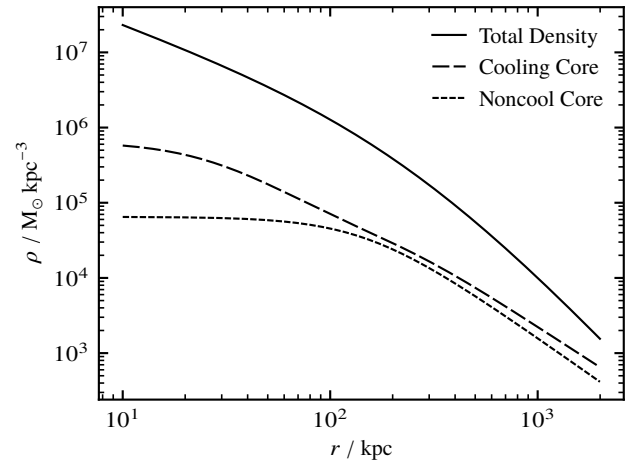


Figure 1: Typical mass density profiles for NCC and CC galaxy clusters are plotted as proxies for all baryonic matter. Shown as well is a standard NFW total density curve that is dominated by a collisionless dark matter component.

Adopting this view, radiative cooling of the higher density inner regions through X-ray emission generally leads to a drop in thermal pressure and a resulting influx of subsonic materials from the surrounding medium, called a cooling flow. This assumption predicts massive rates of molecular condensation near the central galaxies which should produce significant starburst activity. However, observations by the *Chandra* (Weisskopf et al. 2000) and *XMM-Newton* (Jansen et al. 2001) missions proved that this signature is not present in reality. Instead, the ICM cools towards the core but does not reach the neutral gas threshold. The leading hypothesis for the culprit of this deviation from theoretical expectations is the injection of energy by Active Galactic Nucleus (AGN) feedback phases regulated by a balancing of heating and fueling, although this description carries with it its own set of open questions, namely the mechanisms for isotropization of highly directional jets as well as stabilization and dissipation of the AGN bubbles (Böhringer & Werner 2010, Rosati et al. 2002, Pratt et al. 2019).



Figure 2: Selection of galaxy clusters used as examples in this work. Displayed here are composite overlays of second generation *Digitized Sky Survey* red band images and contours for *ROSAT All Sky Survey (RASS)* broad band count rates (Truemper 1982, Lasker 1994). Both optical and X-ray data are stretched using a simple 0.8 exponent power law. Each frame has a radius of $15 \times 10^0'$ for its field of view. Refer to **Figure 8** for raw *ROSAT* counts underlying the contours.

1.2 Classes

According to the spherical dark matter halo assumption, gas properties should follow self similar laws along the radial distance to the cluster center and only scale with the total mass and redshift. Comparing to observations such as those presented in **Figure 7** and **Figure 9** show that this holds quite well for large radii but breaks down closer to the central regions. It turns out that based on these differences, galaxy clusters can be classified into two categories.

(Böhringer & Werner 2010, Pratt et al. 2019)

1.3 Examples

2 Radiation Mechanisms

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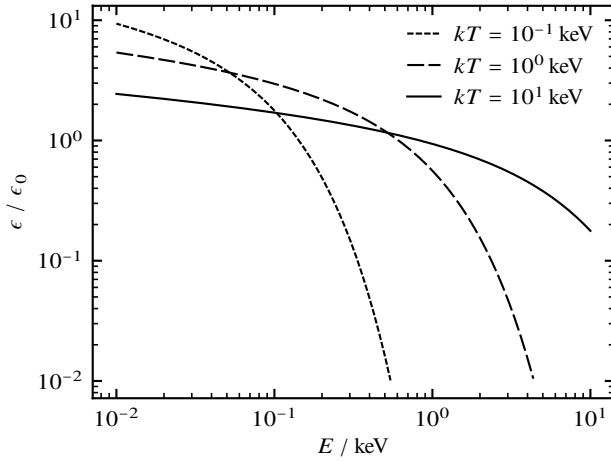


Figure 3: Normalized bremsstrahlung continua for different temperatures over a range of magnitudes. Fitting the bend placement allows for the inference of local temperature.

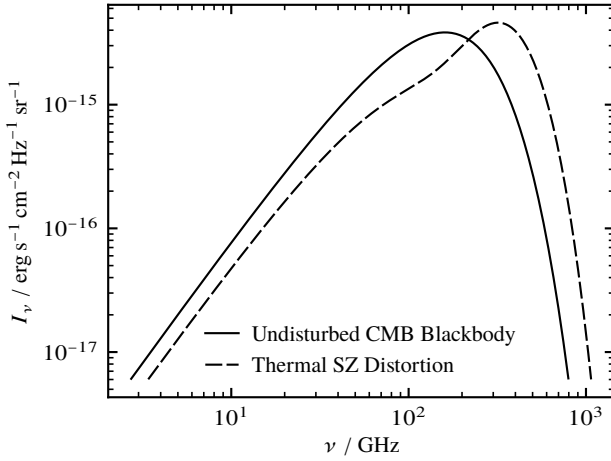


Figure 4: Schematic view of the thermal Sunyaev-Zeldovich effect on the Cosmic Microwave Background (CMB) Planck spectrum with exaggerated $Y \sim 0.2$ Compton parameter in the nonrelativistic limit.

3 Instrumentation

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4 Spectral Characteristics

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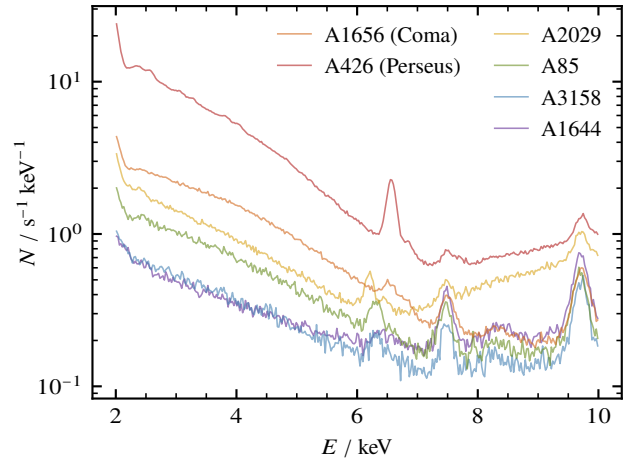


Figure 5: Collection of *Chandra* binned spectra for various galaxy clusters (Weisskopf et al. 2000). Note the presence of emission lines, as well as possible hints of excesses in the relative count rates for both the soft and hard X-ray regimes.

4.1 Continuum

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4.2 Line Emission

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5 Plasma Diagnostics

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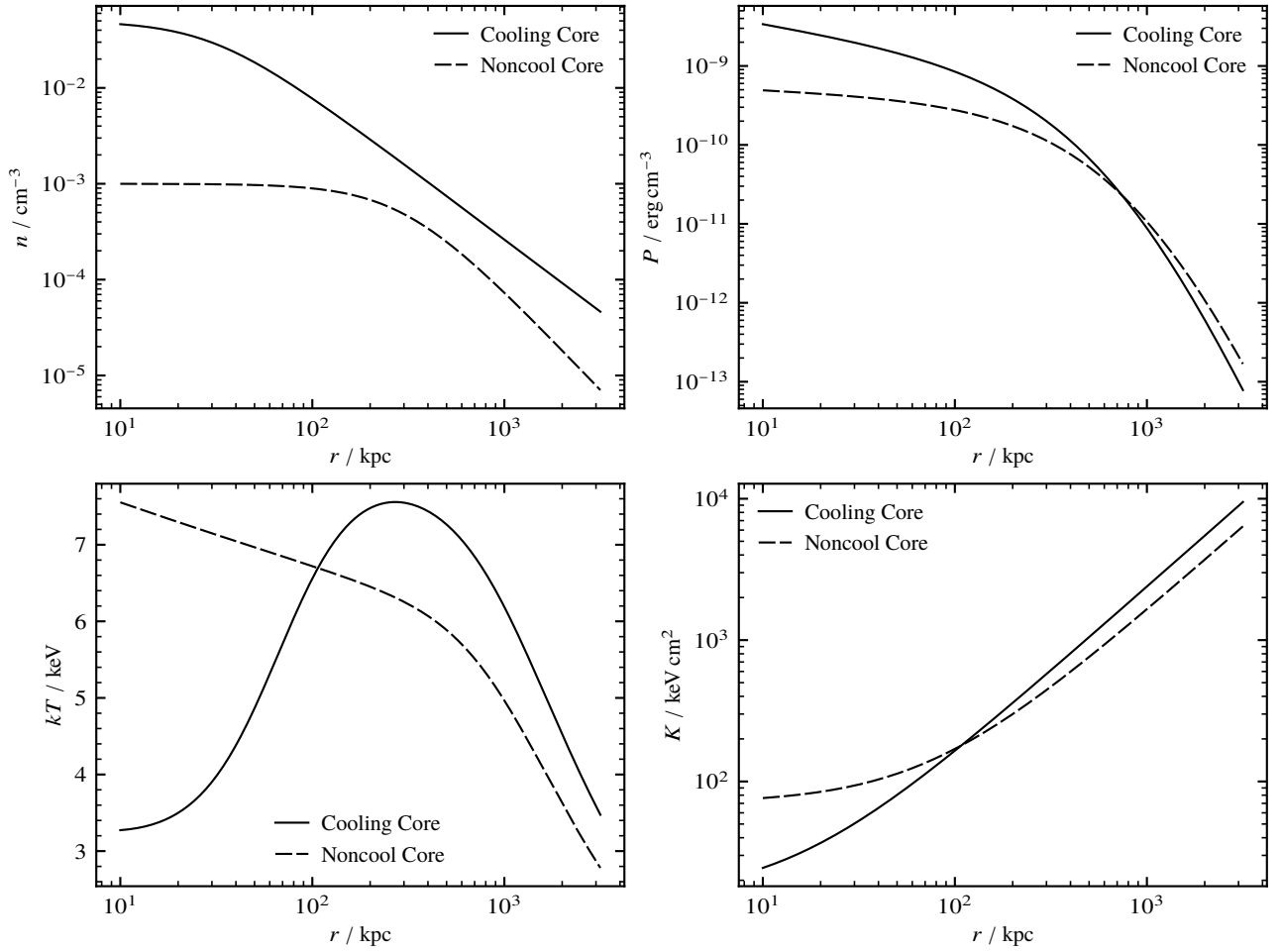


Figure 6: Theoretical galaxy cluster radial profiles assuming universal selfsimilar scaling relations. *Top Left:* Electron number density (Vikhlinin et al. 2006). *Top Right:* Gas pressure (Arnaud et al. 2010). *Bottom Left:* Gas temperature (Vikhlinin et al. 2006). *Bottom Right:* Gas entropy (Cavagnolo et al. 2009). Compare to **Figure 7** for real observations.

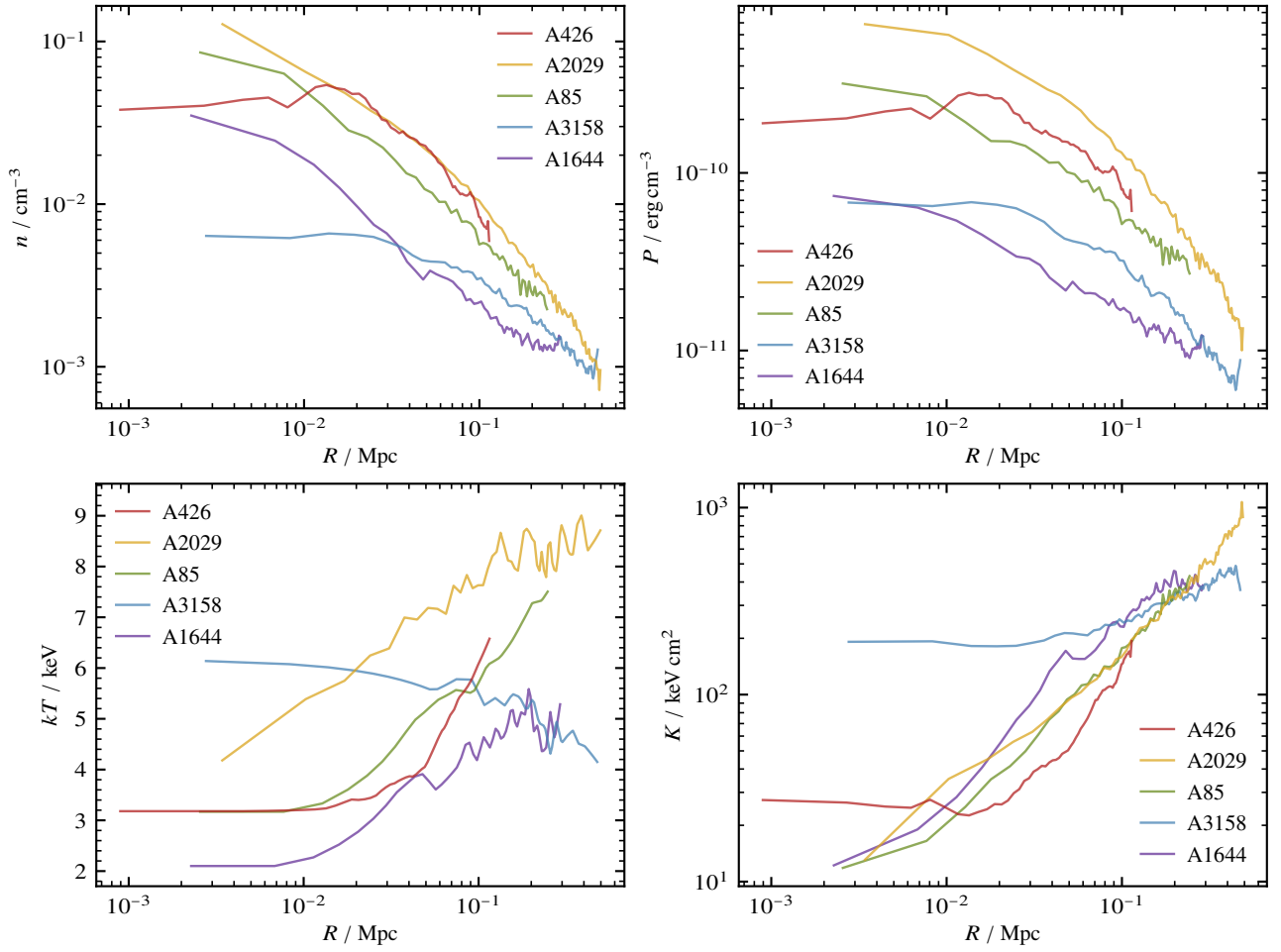


Figure 7: Sample of galaxy cluster radial profiles from *Chandra* observations (Weisskopf et al. 2000). *Top Left:* Electron number density. *Top Right:* Gas pressure. *Bottom Left:* Gas temperature. *Bottom Right:* Gas entropy. Compare to **Figure 6** for a visualization of theoretical and empirical fits with realistic parameter choices.

6 Conclusion

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Appendix

A Reference Images

B Cluster Profiles

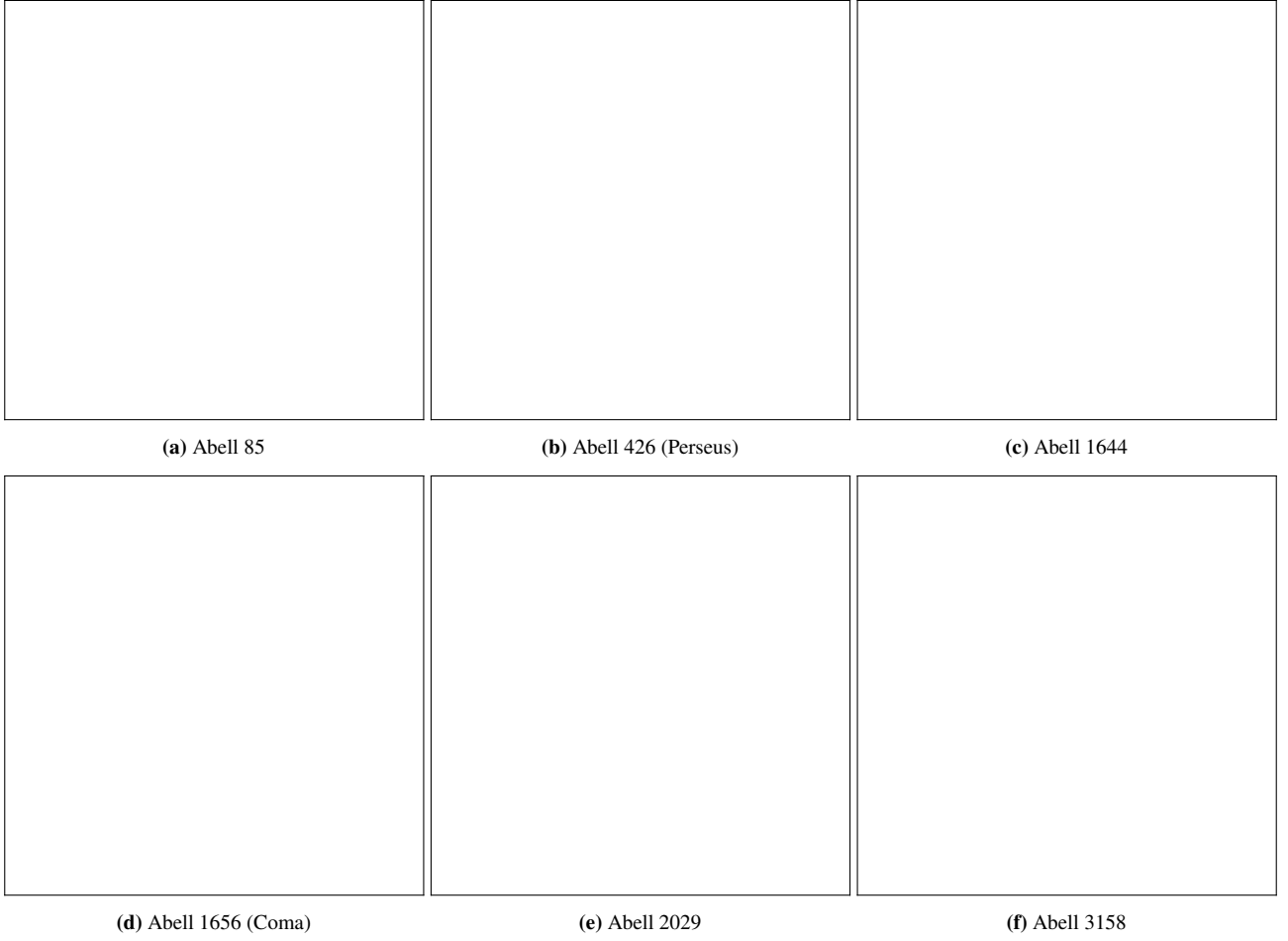


Figure 8: Depicted are *RASS* counts used for X-ray contours shown in **Figure 2** after applying a stretch and Gaussian blur (Truemper 1982). The sampled clusters are selected based on brightness and morphology, as well as data accessibility in the *Archive of Chandra Cluster Entropy Profile Tables (ACCEPT)* and *XMM Cluster Outskirts Project (X-COP)* catalogs (Cavagnolo et al. 2009, Eckert et al. 2017).

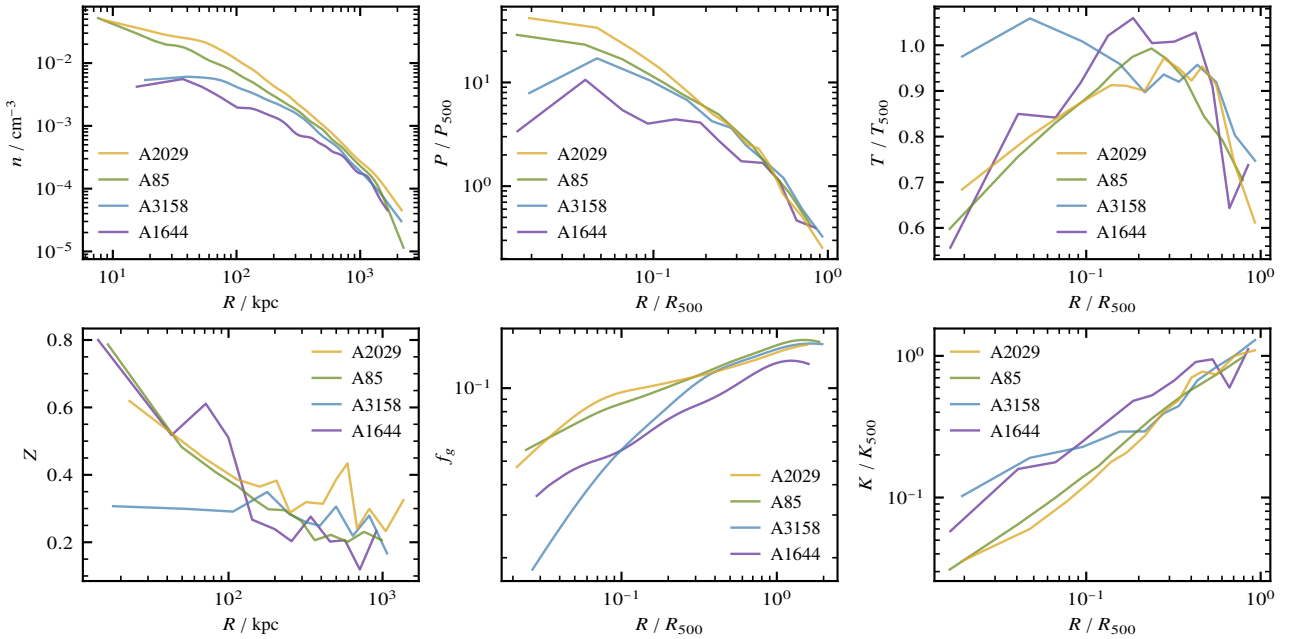


Figure 9: Plots of extracted radial profile projections for available *X-COP* galaxy clusters. *Upper Left:* Electron number density. *Upper Middle:* Gas pressure. *Upper Right:* Gas temperature. *Lower Right:* Gas entropy. *Lower Middle:* Gas mass fraction. *Lower Left:* Metal abundance. Consult the text for definitions of characteristic quantities and scaling. Compare as well to reconstructed distributions in **Figure 7** which focus on the inner regions instead of the cluster outskirts.